



Sri Chaitanya IIT Academy., India.

AP, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office, Madhapur – Hyderabad

Sec: Sr-ICON-B-1

JEE-ADVANCE-2017-P1

Date: 16-05-21

Time: 07:00 AM to 10:00 AM

GTA-04

Max.Marks: 183

SYLLABUS:

Physics : TOTAL SYLLABUS

Chemistry : TOTAL SYLLABUS

Mathematics : TOTAL SYLLABUS

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2017-P1-Model****Time: 03:00 Hr's****IMPORTANT INSTRUCTIONS****Max Marks: 183****PHYSICS:**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 7)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 8 – 12)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 13 – 18)	Questions With Single Answer Type	+3	-1	6	18
Total				18	61

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 25)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 26 – 30)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 31 – 36)	Questions With Single Answer Type	+3	-1	6	18
Total				18	61

MATHEMATICS:

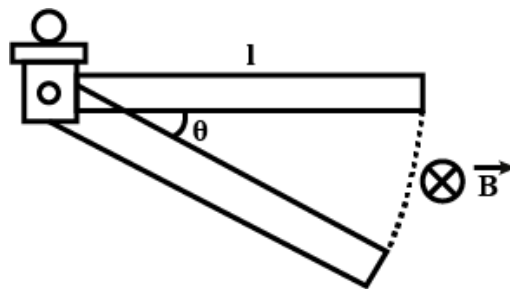
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 43)	Questions With Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	7	28
Sec – II (Q.N : 44 – 48)	Questions With Integer Answer Type	+3	0	5	15
Sec – III (Q.N : 49 – 54)	Questions With Single Answer Type	+3	-1	6	28
Total				18	61

**PHYSICS****Max. Marks:61****SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)**

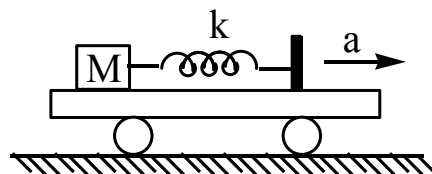
This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

01. $y(x, t) = 0.8 / [(4x + 5t)^2 + 5]$ represents a moving pulse, where x and y are in meter and t in second. Then
- A) pulse is moving in $+x$ direction
B) in 2s it will travel a distance of 2.5 m
C) its maximum displacement is 0.16m
D) it is a symmetric pulse
02. A conducting rod of length l is hinged at point O. It is free to rotate in a vertical plane. There exists a uniform magnetic field B in horizontal direction. The rod is released from the position shown. The potential difference between the two ends of the rod is proportional to



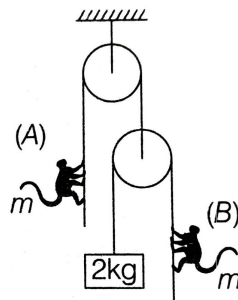
- A) $l^{3/2}$ B) l^2 C) $\sin \theta$ D) $(\sin \theta)^{1/2}$
03. A block of mass M is attached with a spring of spring constant k . The whole arrangement is placed on a vehicle as shown in the figure. If the vehicle starts moving towards right with an acceleration a (there is no friction anywhere), then



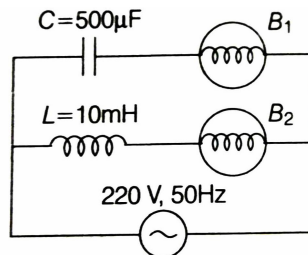


- A) maximum elongation in the spring is $\frac{Ma}{k}$
- B) maximum elongation in the spring is $\frac{2Ma}{k}$
- C) maximum compression in the spring is $\frac{2Ma}{k}$
- D) maximum compression in the spring is zero

04. Two monkeys (A) and (B) of same mass $m = 1\text{kg}$ are hanging on the strings such that block of mass 2kg remains at rest and it is given that monkey (B) is just holding the string. Then which of the following statement(s) is/are correct ($g = 10\text{m/s}^2$):

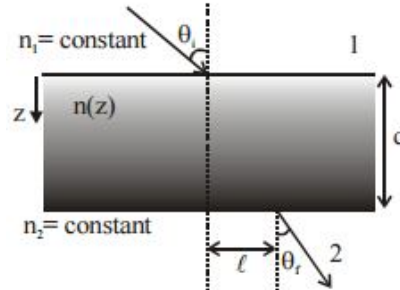


- A) Acceleration of monkey (B) is 10m/s^2 upwards
- B) Acceleration of monkey (A) is 30m/s^2 upwards
- C) Acceleration of monkey (A) with respect to his rope is 35m/s^2 upwards
- D) Acceleration of monkey (B) with respect to his rope is 50m/s^2 upwards
05. In the circuit shown in figure, if both the bulbs B_1 and B_2 are identical



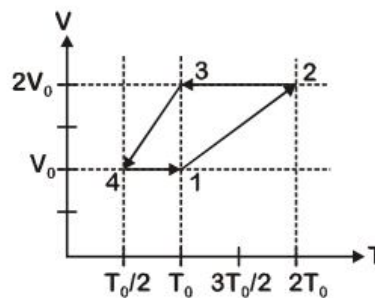
- A) their brightness will be the same
- B) B_2 will be brighter than B_1
- C) as frequency of supply voltage is increased the brightness of bulb B_1 will increase and that of B_2 will decrease
- D) Brightness of both bulbs is independent of the frequency

- 06.** A transparent of thickness d has a refractive index $n(z)$ that increases with z . Here z is the vertical distance inside the slab, measured from the top. The slab is placed between two media with uniform refractive indices n_1 and $n_2 (> n_1)$, as shown in the figure. A ray of light is incident with angle θ_i , from medium 1 and emerges in medium 2 with refraction angle θ_t with a lateral displacement l .



Which of the following states(s) is (are) true ?

- A)** $n_1 \sin \theta_i = n_2 \sin \theta_f$
B) $n_1 \sin \theta_i = (n_2 - n_1) \sin \theta_f$
C) l is independent of n_2
D) l is dependent on $n(z)$
- 07.** One mole of a monatomic ideal gas goes through a thermodynamic cycle, as shown in the volume versus temperature ($V - T$) diagram. The correct statement(s) is/are : [R is the gas constant]



- A)** Work done in this thermodynamic cycle ($1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$) is $|W| = \frac{1}{2}RT_0$
- B)** The above thermodynamic cycle exhibits only isochoric and adiabatic processes.
- C)** The ratio of heat transfer during processes $1 \rightarrow 2$ and $2 \rightarrow 3$ is $\left| \frac{Q_{1 \rightarrow 2}}{Q_{2 \rightarrow 3}} \right| = \frac{5}{3}$
- D)** The ratio of heat transfer during processes $1 \rightarrow 2$ and $3 \rightarrow 4$ is $\left| \frac{Q_{1 \rightarrow 2}}{Q_{3 \rightarrow 4}} \right| = \frac{1}{2}$

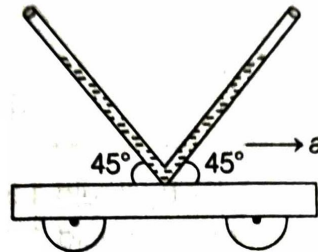


SECTION-II
(INTEGER ANSWER TYPE)

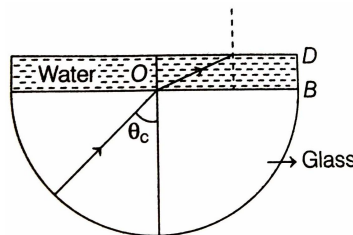
This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

- 08.** A bent tube of uniform cross section is mounted on a cart which is accelerating towards right with constant acceleration 'a'. The total length of the liquid is $2\sqrt{2}l$. Find the level difference between two limbs in equilibrium ($a = 2m/s^2, l = 10m$)



- 09.** A ray of light travelling in glass ($\mu = 3/2$) is incident on a horizontal glass air surface at the critical angle θ_c . If a thin layer of water ($\mu = 4/3$) is now poured on the glass air surface, the ray of light emerge into air at the water air surface at an angle of π/k , radians find the value of k.



- 10.** A H-atom moving with speed v makes a head on collision with a H-atom at rest. Both atoms are in ground state. The minimum value of velocity v for which one of the atom may excite is $(1.25n \times 10^4)m/s$. Find value of n .
- 11.** The magnetic field at the centre of a hydrogen atom due to the motion of the electron in the first Bohr orbit is B . The magnetic field at the centre due to the motion of the electron in the second Bohr orbit will be $\frac{B}{2^x}$. Find value of x .



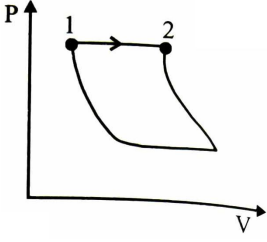
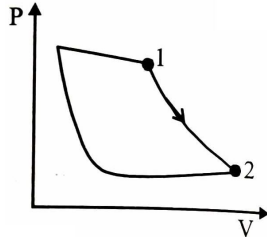
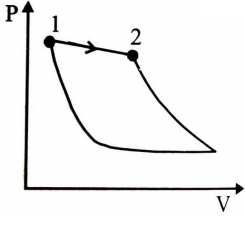
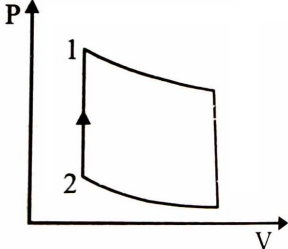
12. A train of sound waves is propagated along a wide pipe and it is reflected from an open end. If the amplitude of the waves is 0.002 cm, the frequency 1000 Hz and the wavelength 40 cm, the amplitude of vibration at a point 10 cm from open end inside the pipe will be (0.0x) cm. Find value of x.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.13, Q.14 and Q.15 by appropriately matching the information given in the three columns of the following table.

Column – I	Column – II	Column – III
(I) $W_{1 \rightarrow 2} = \frac{1}{\gamma - 1} (P_2 V_2 - P_1 V_1)$	(i) Isothermal	(P) 
(II) $W_{1 \rightarrow 2} = -PV_2 + PV_1$	(ii) Isochoric	(Q) 
(III) $W_{1 \rightarrow 2} = 0$	(iii) Isobaric	(R) 
(IV) $W_{1 \rightarrow 2} = -nRT \ln \left(\frac{V_2}{V_1} \right)$	(iv) Adiabatic	(S) 



13. Which of the following options correctly represents a thermodynamic process that is used as a correction in the determination of the speed of sound in an ideal gas ?
 A) (I) (ii) (Q) B) (IV) (ii) (R) C) (III) (iv) (R) D) (I) (iv) (Q)
14. Which one of the following options is the correct combination ?
 A) (IV) (ii) (S) B) (III) (ii) (S) C) (II) (iv) (P) D) (II) (iv) (R)
15. Which of the following options is the only correct representation of a process in which $\Delta U = \Delta Q - P\Delta V$?
 A) (II) (iv) (R) B) (III) (ii) (P) C) (II) (iii) (S) D) (II) (iii) (P)

Answer Q.16, Q.17 and Q.18 by appropriately matching the information given in the three columns of the following table.

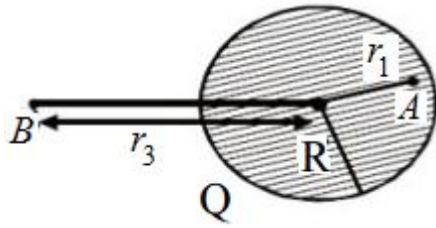
Column-I & II represents the magnitude of electric field at points A & B respectively. Column- III represents the magnitude of potential difference

between the two points. (Take potential at infinity to be zero) $\left(k = \frac{1}{4\pi\epsilon_0} \right)$

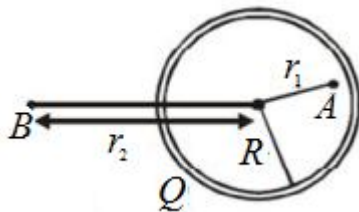
Column – I Electric field at point A	Column – II Electric field at point B	Column – III Potential difference between two points
(I) $\frac{\rho r_1}{2\epsilon_0}$	(i) $\frac{\rho R^2}{4\epsilon_0 r_2}$	(P) $ V_A - V_B = \frac{kQ}{2R} \left(3 - \frac{r_1^2}{R^2} \right) - \frac{kQ}{r_2}$
(II) $\frac{kQr_1}{R^3}$	(ii) $\frac{\rho R^2}{2\epsilon_0 r_2}$	(Q) $ V_A - V_B = \frac{3kQ}{2R} - \frac{kQ}{r_2}$
(III) 0	(iii) $\frac{kQ}{r_2^2}$	(R) $ V_C - V_B = \frac{\rho R^2}{2\epsilon_0} \ln \left(\frac{r_3}{r_2} \right)$
(IV) $\frac{\rho r_1}{4\epsilon_0}$	(iv) $\frac{kQr_2}{R^3}$	(S) $ V_A - V_B = \frac{kQ}{R} - \frac{kQ}{r_2}$



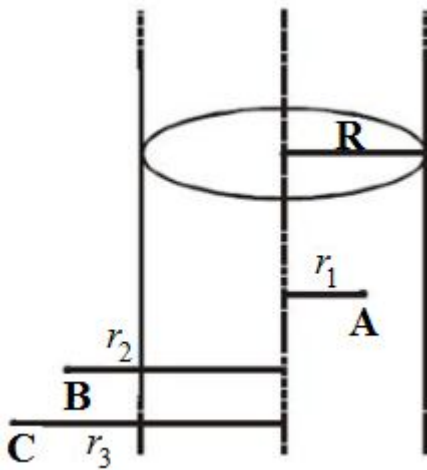
16. Which of the following combination represents the situation due to a uniformly charged non-conducting solid sphere having total charge Q & radius R :-



- A) (II) (iii) (Q) B) (III) (iii) (S) C) (II) (iii) (P) D) (III) (iv) (P)
17. Which of the following combination represents the situation due to a uniformly charged non-conducting hollow thin sphere having total charge Q & radius R :-



- A) (III) (iii) (S) B) (II) (iii) (P) C) (III) (iv) (Q) D) (II) (iv) (S)
18. Which of the following combination represent the situation due to a uniformly charged infinite solid cylinder of radius R and charge density ρ



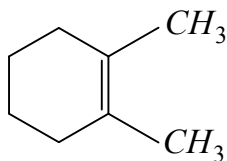
- A) (III) (i) (P) B) (I) (i) (R) C) (IV) (ii) (P) D) (I) (ii) (R)

**CHEMISTRY****Max.Marks:61****SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)**

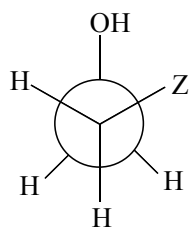
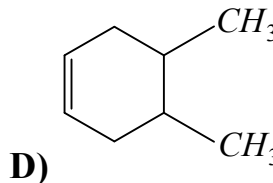
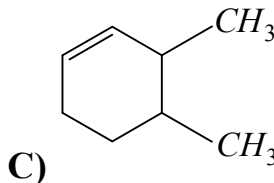
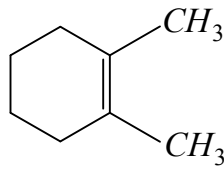
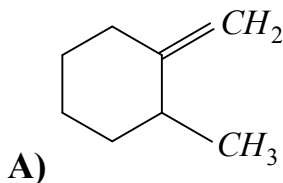
This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

19. Choose the correct statements :
- A) Temperature, enthalpy and entropy are state functions
- B) For both, reversible and irreversible isothermal expansion of an ideal gas, change in internal energy and enthalpy is zero
- C) For a reaction in which $\Delta n_g = 0$, entropy change is not always zero
- D) The entropy change associated with reversible isothermal expansion of an ideal gas is equal to $2.303R \log_{10} \frac{P_1}{P_2}$
20. Benzene and naphthalene form an ideal solution at room temperature. For this process, the true statement(s) is (are) :
- A) ΔG is positive
- B) ΔS_{system} is positive
- C) $\Delta S_{surroundings} = 0$
- D) $\Delta H = 0$



21. Heat of hydrogenation of _____ is less than :



22. The above Gauche form is stable when Z is

- A) $-F$ B) $-OH$ C) $-C_2H_5$ D) $-CH_3$



23. The correct statement(s) regarding
i) $HClO$ ii) $HClO_2$ iii) $HClO_3$ and iv) $HClO_4$
is (are) :
- A) The number of $Cl = O$ bonds in (ii) and (iii) together is two
B) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
C) The hybridisation of Cl in (iv) is sp^3
D) Amongst (i) to (iv), the strongest acid is (i)
24. Which one of the following arrangements does not give the correct picture of the trends indicated against it ?
- A) $F_2 > Cl_2 > Br_2 > I_2$: Bond dissociation energy
B) $F_2 > Cl_2 > Br_2 > I_2$: Electro negativity
C) $F_2 > Cl_2 > Br_2 > I_2$: Oxidising nature
D) $F_2 > Cl_2 > Br_2 > I_2$: Electron gain enthalpy
25. Which of the following are outer orbital complexes ?
- A) $[Co(NH_3)_6]^{3+}$ B) $[Mn(CN)_6]^{3-}$ C) $[Ni(NH_3)_6]^{2+}$ D) $[FeF_6]^{3-}$

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

26. In lead storage battery, how many moles of electrons will be involved in the consumption of 98g of H_2SO_4 ?
27. Ice crystallizes in hexagonal lattice having volume of the unit cell is $132 \times 10^{-24} cm^3$. If the density of ice is $0.92 g cm^{-3}$ at a given temperature, then number of H_2O molecules per unit cell is
28. Total number of sigma bonds in the following compounds according to molecular orbital theory. $He_2, Li_2, B_2, C_2, N_2, O_2, F_2$



29. Out of I_3^- , ICl_2^- , $BeCl_2$, XeF_2 , XeF_6 , BrF_5 , SCl_2 , NO_2^+ . How many will have linear shape ?
30. Total number of alkenes possible by dehydrobromination of 3 – bromo – 3 – cyclopentyl hexane using alcoholic KOH is

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.31, Q.32 and Q.33 by appropriately matching the information given in the three columns of the following table.

Column – I	Column – II	Column – III
A) H – atom	P) Radius of 2 nd orbit is $0.53 \times 2A^0$	i) Energy of the 1 st excited state is –13.6 eV
B) He^+ ion	Q) Energy of the 4 th orbit is –13.6 eV	ii) Radius of 2 nd orbit is $0.53A^0$
C) Be^{+3} ion	R) Radius of the 3 rd orbit is $0.53 \times 3A^0$	iii) Velocity of 1 st energy level is $2.188 \times 10^8 \times 3 \text{ cm / sec}$
D) Li^{+2} ion	S) Velocity of e^- in the 1 st orbit is $2.18 \times 10^8 \text{ cm / sec}$	iv) Velocity of the electron in the 3 rd orbit is $2.18 \times 10^3 \times \frac{1}{3} \text{ cm / sec}$

31. For He^+ ion the correct combination is
 A) (B) (P) (i) B) (B) (P) (ii) C) (B) (R) (ii) D) (B) (Q) (i)
32. In case of Li^{+2} ion, the correct combination is
 A) (D) (s) (II) B) (D) (P) (iv) C) (D) (Q) (iii) D) (D) (R) (iii)
33. For H – atom, the correct combination is
 A) (A) (S) (ii) B) (A) (S) (iv) C) (A) (P) (iv) D) (A) (Q) (iv)



Answer Q.34, Q.35 and Q.36 by appropriately matching the information given in the three columns of the following table.

Column – I (reaction)	Column – II (reactant species)	Column – III (condition)
I) Hofmann	i) $\begin{array}{c} O \\ \\ R - C - NH - OH \end{array}$ (Hydroxamic acid)	P) CrO_2Cl_2 / CS_2
II) Lossen	ii) Toluene	Q) $Br_2 / OH^\ominus$
III) Reimer – Tiemann	iii) $\begin{array}{c} O \\ \\ R - C - NH_2 \end{array}$	R) $CHCl_3 / OH^\ominus$
IV) Etard	iv) Phenol	S) $OH^\ominus$

34. For Hofmann reaction, the correct combination is

- A) (I) (iv) (S) B) (I) (ii) (R) C) (I) (iii) (S) D) (I) (iii) (Q)

35. For Reimer – Tiemann reaction, the correct combination is

- A) (III) (iv) (P) B) (III) (iv) (R) C) (III) (i) (S) D) (III) (ii) (Q)

36. For Etard reaction, the correct combination is

- A) (IV) (ii) (P) B) (IV) (i) (P) C) (IV) (iii) (Q) D) (IV) (iii) (S)

**MATHEMATICS****Max.Marks:61****SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

37. In a right angled triangle the lengths of the sides are 'a' and 'b' ($0 < a < b$). A circle passes through the mid point of the smaller side and touches the hypotenuse at it's mid point
- A) Center of circle $\left(\frac{2a^2 - b^2}{4a}, \frac{b}{4}\right)$ B) Center of circle $\left(\frac{a^2 - b^2}{4a}, \frac{b^2}{4}\right)$
- C) radius of circle is $\frac{b\sqrt{a^2 + b^2}}{4a}$ D) radius of circle is $\frac{a\sqrt{a^2 + b^2}}{4a}$
38. Ram rolls a fair die until 6 appears first time. Som also rolls a fair die independently until 5 appears first time. Then
- A) Probability that ram gets 6 as result on third throw is $\frac{25}{216}$
- B) Probability that Ram gets 6 as a result on or after third throw $\frac{25}{36}$
- C) Probability that Som gets 5 as a result on or after third throw is $\frac{25}{211}$
- D) Let α and β be number of throws respectively required by Ram and Som to obtain 6 and 5, then probability that $(\alpha - \beta) < 1$ is $\frac{8}{33}$
39. If a complex number Z satisfies equation $|Z| - \log_2(Z + 2) = 0, (\text{Re } Z > 0)$ then
- A) $\text{Arg}Z = \frac{\pi}{2}$ B) $|Z| = 2$ C) $\text{Arg}Z = 0$ D) $Z = \bar{Z}$
40. $\cos \frac{\pi}{7}, \cos \frac{3\pi}{7}, \cos \frac{5\pi}{7}$ are roots of equation $8x^3 - 4x^2 - 4x + 1 = 0$ then
- A) $\sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7} = 4$ B) $\sec \frac{\pi}{7} \times \sec \frac{3\pi}{7} \times \sec \frac{5\pi}{7} = -8$
- C) $\sin \frac{\pi}{14} \times \sin \frac{3\pi}{14} \times \sin \frac{5\pi}{14} = \frac{1}{8}$ D) $\cos \frac{\pi}{14} \times \cos \frac{3\pi}{14} \times \cos \frac{5\pi}{14} = \frac{\sqrt{7}}{8}$



41. If α and β are real values of x which satisfy the equation

$$\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x \text{ then}$$

- A) $\alpha + \beta = \frac{1}{2}$ B) $\alpha.\beta = 0$ C) $|\alpha - \beta| = \frac{1}{2}$ D) $\alpha + \beta = 1$

42. If p, q, r are positive integers and $\Delta = 12 \begin{vmatrix} pc_1 & pc_2 & pc_3 \\ qc_1 & qc_2 & qc_3 \\ rc_1 & rc_2 & rc_3 \end{vmatrix}$ then

- A) Δ is even integer B) Δ is divisible by 1
C) Δ is a rational number D) Δ is an odd integer

43. For the hyperbola $\frac{x^2}{9} - \frac{y^2}{3} = 1$ in correct statement is

- A) the acute angle between it's asymptotes is 60°
B) it's eccentricity is $\frac{4}{3}$
C) length of latusrectum is 2
D) product of perpendicular from any point on the hyperbola to it's asymptotes is $\frac{9}{4}$

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

44. If an extreme value of $f(x) = \int_0^x (t-1)(t-2)dt$ is k , then $6k =$

45. If the sum $\sum_{k=1}^{\infty} \frac{1}{(k+2)\sqrt{k} + k\sqrt{k+2}} = \frac{\sqrt{a} + \sqrt{b}}{\sqrt{c}}$ where $a, b, c \in N$ then $a + c - b =$

46. The number of values of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ satisfying the equation

$$(1 - \tan \theta)(1 + \tan \theta) \sec^2 \theta + 2^{\tan^2 \theta} = 0$$



47. The general solution of $(1 + e^{x/y})dx + e^{x/y}\left(1 - \frac{x}{y}\right)dy = 0$ is a curve passing through $(0, 1)$,

then $x + ye^{x/y} =$

48. Let $\vec{a} = \vec{i} + \vec{j} + \vec{k}, \vec{b} = \vec{i} - \vec{j}$ then the vectors $\vec{\alpha} = (\vec{a} \cdot \vec{i})\vec{i} + (\vec{a} \cdot \vec{j})\vec{j} + (\vec{a} \cdot \vec{k})\vec{k}$,

$\vec{\beta} = (\vec{b} \cdot \vec{i})\vec{i} + (\vec{b} \cdot \vec{j})\vec{j} + (\vec{b} \cdot \vec{k})\vec{k}$ and $\vec{\gamma} = \vec{i} + \vec{j} - 2\vec{k}$ form a parallelepiped of volume

SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.49, Q.50 and Q.51 by appropriately matching the information given in the three columns of the following table.

Column-1 contains functions, Column-2 contains domain and Column-3 contains ranges.

Column – 1	Column – 2	Column – 3
(I) $\sqrt{\cos(\sin x)}$	(i) $[1, 2]$	(P) $\left[0, \frac{\pi}{2}\right]$
(II) $\sqrt{\sin(\cos x)}$	(ii) $\left[2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{2}\right], n \in \mathbb{Z}$	(Q) $\mathbb{R}$
(III) $\sqrt{\sin^{-1}(\log_2^x)}$	(iii) $(1, 2)$	(R) $[0, \sqrt{\sin 1}]$
(IV) $\log\left(\frac{\sqrt{4-x^2}}{1-x}\right)$	(iv) $\mathbb{R}$	(S) $[\sqrt{\cos 1}, 1]$

49. Which of the following is only the correct combination

A) (I) (ii) (P) **B)** (II) (iv) (R) **C)** (III) (i) (P) **D)** (IV) (iii) (Q)

50. Which of the following is the only incorrect combination

A) (I) (iv) (S) **B)** (II) (ii) (R) **C)** (IV) (i) (P) **D)** (III) (i) (P)

51. Which of the following is only the correct combinations

A) (II) (ii) (S) **B)** (II) (ii) (R) **C)** (III) (iv) (P) **D)** (I) (iii) (Q)



Answer Q.52, Q.53 and Q.54 by appropriately matching the information given in the three columns of the following table.

Column-1, 2 & 3 contain parameters equation of conics, equation of chord joining two parametric point and equation of tangents at parametric point respectively.

Column – 1	Column – 2	Column – 3
(I) $x = 2 \cos \theta$, $y = a \sin \theta$	(i) $\frac{x}{2} \cos \left(\frac{\theta_1 - \theta_2}{2} \right) - \frac{y}{a} \sin \left(\frac{\theta_1 + \theta_2}{2} \right) = \cos \left(\frac{\theta_1 + \theta_2}{2} \right)$	(P) $\frac{x}{2} - \frac{y}{a} \sin \theta = \cos \theta$
(II) $x = 2 \sec \theta$, $y = a \tan \theta$	(ii) $\frac{x}{2} \sin \left(\frac{\theta_1 + \theta_2}{2} \right) - \frac{y}{a} \cos \left(\frac{\theta_1 - \theta_2}{2} \right) + \cos \left(\frac{\theta_1 + \theta_2}{2} \right) = 0$	(Q) $y \sin 2\theta$ $= 2x \sin^2 \theta + 2a \cos^2 \theta$
(III) $x = 2 \tan \theta$, $y = a \sec \theta$	(iii) $y \sin(\theta_1 + \theta_2)$ $= 2x \sin \theta_1 \sin \theta_2 + 2a \cos \theta_1 \cos \theta_2$	(R) $\frac{x}{2} \cos \theta + \frac{y}{a} \sin \theta = 1$
(IV) $x = a \cot^2 \theta$, $y = 2a \cot \theta$	(iv) $\frac{x}{2} \cos \left(\frac{\theta_1 + \theta_2}{2} \right) + \frac{y}{a} \sin \left(\frac{\theta_1 + \theta_2}{2} \right) = \cos \left(\frac{\theta_1 - \theta_2}{2} \right)$	(S) $\frac{x}{2} \sin \theta - \frac{y}{a} + \cos \theta = 0$

52. For $a = 2$ if a tangent is drawn to a suitable conic whose equation is $x - 3y + 18 = 0$, then the correct combination of that conic is:

A) (II) (ii) (S) **B)** (III) (ii) (Q) **C)** (I) (ii) (R) **D)** (IV) (iii) (Q)

53. Which of the following is a correct combination ?

A) (III) (i) (S) **B)** (II) (i) (P) **C)** (IV) (ii) (Q) **D)** (I) (ii) (R)

54. Which of the following is a correct combination ?

A) (III) (ii) (S) **B)** (II) (iv) (P) **C)** (IV) (i) (Q) **D)** (I) (iii) (R)